

ASSESSMENT AND INTERNAL VERIFICATION FRONT SHEET (Individual Criteria)

Course Title	Advanced Diploma in IT (MQF Level 4)			Lecturer Name & Surname	Ing. James Attard / Mr. James Decelis / Mr. Victor John Saliba / Mr. Kris Domancich	
Unit Number & Title		ITSFT-406-2001 Introduction to Mobile Applications Development				
Assignment Number, Title / Type		02, Create a Functional Ionic-React App / Home Assignment				
Date Set		20/04/2026	Deadline Date	15/05/2026		
Student Name			ID Number		Class / Group	

Assessment Criteria	Maximum Mark
<i>KU1.5: Outline the strengths and weaknesses of mobile technologies</i>	5
<i>KU2.1: Explain the web/mobile application's functional requirements</i>	5
<i>KU3.5: Identify potential improvements for the prototype application</i>	5
<i>AA3.3: Examine the core functionalities of the built prototype</i>	7
<i>AA3.4: Develop a suitable application that satisfies the given scenario</i>	7
<i>SE1.3: Create an application capable of running on a mobile platform using a given technology</i>	10
<i>SE1.4: Evaluate the strengths and weaknesses of a mobile application when compared to another technology</i>	10
Total Mark	49

Notes to Students:
- This assignment brief has been approved and released by the Internal Verifier through Classter. - Assessment marks and feedback by the lecturer will be available online via Classter (http://mcast.classter.com) following release by the Internal Verifier - Students submitting their assignment on Moodle/Turnitin will be requested to confirm online the following statements: Student's declaration prior to handing-in of assignment ❖ I certify that the work submitted for this assignment is my own and that I have read and understood the respective Plagiarism Policy Student's declaration on assessment special arrangements ❖ I certify that adequate support was given to me during the assignment through the Institute and/or the Inclusive Education Unit. ❖ I declare that I refused the special support offered by the Institute.

Create a Functional Ionic-React Application

Assignment Guidelines

- This is a **Home Assignment** in the form of a fully functional mobile application project.
- Fill in **all the details in the cover sheet** (first page of this document) otherwise your submission will not be accepted.
- **All students will be required to do a brief class presentation during the designated lecture slot to explain their work in front of the other students.**
- The project consists of two submissions – the proposal and the project.
- Your first submission is your project proposal (as per requirement 1). Submission method: **Soft copy** on **VLE**. You should upload a document that contains the following:
 1. Your name and the title of your mobile application with a one paragraph description of its proposed features.
 2. At least two Screenshots/Diagrams of your proposed application.
- Your second submission should be the actual project. Submission method: **Soft copy** on **VLE**. You should upload **two files** – a compressed folder and this document in its entirety, filled with your details. These two files need to be uploaded to VLE using your **class submission link**
 1. **First File - Compressed Folder:**
 - Please **DO NOT INCLUDE THE ENTIRE IONIC PROJECT** in this folder. Add **only** the following files:
 - The source code files: **App.tsx** and the **/pages** folder which will contain your application screens/pages
 - Custom Icon used in app deployment
 - Link to your short Video Demonstration
 - **Compress** the folder (zip or rar) and call it with **your** name, surname and class (eg: *Joe_Borg_SW42A*). Finally, **upload** the compressed file
 2. **Second File – The Cover Sheet filled with your details:**
 - In addition to the previous compressed folder, you need to upload the cover sheet with your details. You can copy the cover sheet from the first page of this document.

Read Carefully

- Submissions occurring after the indicated time frame are considered late and thus will not be accepted.
- All the institute procedures, rules and regulations apply to this assignment.
- **Copying and unethical use of AI tools is strictly prohibited and will be penalized** in line with the College's disciplinary procedures.

ASSIGNMENT OVERVIEW

In this assignment you are tasked with building a mobile app using Ionic-React. You may choose any theme (e.g., productivity, fitness, study planner, budget tracker, hobby tracker, etc.), but your application must include the following:

- Appropriate Ionic components and styling
- Implement navigation (tabs, side menu, or multi-page routing) to allow the user to switch between at least two pages
- Use React state (useState) to manage data
- Allow users to:
 - Add data
 - Display data
 - Update OR delete data
- Add data persistency features
- Add data validation features
- Successfully deploy and run the app using Capacitor on Android Emulator

Within 1 week from the assignment issue date you are required to submit a proposal on VLE consisting of the following:

- Title of your proposed application;
- A paragraph describing the theme you chose for your mobile app;
- At least x2 wireframe designs, one for each screen of the mobile app.

Note: Before you create the proposal, it is suggested to read the entire assignment sheet to formulate a better prototype proposal.

For examples of app scenarios, refer to the bottom of this document.

Requirement 1 – Proposal (KU2.1 - 2 marks)

Your goal in this section is to submit a proposal on VLE within 1 week from the assignment issue date. You will be asked to present the proposal in the classroom during the lecture in front of other students. The proposal should be in a word document consisting of the following:

- Title of your proposed application;
- A paragraph describing the theme you chose for your mobile app;
- At least x2 wireframe designs, one for each screen of the mobile app.

A comprehensive list of examples of app scenarios can be found towards the bottom of this document.

Note: Before you create the proposal, it is suggested to read the entire assignment sheet to formulate a better prototype proposal.

Requirement 2 – App Development in Ionic-React (KU2.1, AA3.4, SE1.3, SE1.4 - 30 marks)

Your goal in this section is to develop a fully functional mobile app based on your submitted proposal. The following is a list of all the specifications required to build this application. Complete as many of these challenges as possible to get the most marks!

1. Navigation Menu (3 marks)

The application needs to have at least two screens which can be switched through a navigation menu of your choice.

2. User Interface (2 marks)

User interface should be developed using only Ionic Components - any styling should be limited to Ionic styling as well. It is up to you to decide the best Ionic Components to use.

3. Core React Functionality (10 marks)

The app should use React to allow a user to add, view and delete/update data. Suitable UI components (such as buttons, sliders, etc) should be used to implement this functionality.

4. Validation (3 marks)

The application should not allow the user to submit an empty input. The application should also confirm with the user prior to deleting an activity.

5. Activity Completion (6 marks)

Suitable UI components (such as counters, checkboxes, range sliders, strikethrough styling, etc) should be implemented to track user's activities including but not limited to completion, deletion, addition, and other data related activities.

6. Persisting Data (6 marks)

All data in the application should persist when the browser/emulator is either re-opened or refreshed. This includes but is not limited to, data setlist, activity tracking etc. Please note that Data (task 3) and Activity Completion (task 5) must be persisted as well.

Requirement 3 – Deployment Strategy (KU1.5, KU3.5 - 10 marks)

Your task in this section is to demonstrate your understanding of deploying a mobile application in the **real world**. As such, you are asked to:

1. Deploy this application to an **Android emulator** using Android Studio and Capacitor.

2. Design a new icon for the app, using any tool of your choice.
3. Do some research to deploy this application on the emulator using your custom icon in the previous step, as long as it is not the default Capacitor icon.

To demonstrate that you have fulfilled this requirement, you need to include this in your video presentation (refer to **Requirement 4**) by opening the emulator of your choice, showing the application icon, and running the application from the emulator.

Requirement 4 – Video & Class Presentation (AA3.3 - 7 marks)

Create a short video showing a demo of your application. The demo should include both the screen recording and your own voiceover explaining the demo and parts of the code. More details below:

- 1) The application needs to be run on an Android Studio Emulator of your choice (if you have completed the task as per **Requirement 3**) or else *you can use a regular browser if you have not fulfilled Requirement 3*.
- 2) Your demo should include how your data is being persisted (as per **Requirement 2.6**). If you are using the browser to demonstrate your application, make sure you record the entire browser screen so you can demonstrate that data persists even when you click the browser's refresh. If you are using an Emulator, you need to close the application and open it again from the Emulator, or else you may also close the entire Emulator and reopen it again to demonstrate that your data persists.
- 3) Apart from a demo of the deployed application, the video should also show code snippets of your application where you will explain the **key parts of the code** (especially **Requirements 2.3 – 2.6**)
- 4) The video should not be longer than **5 minutes** and should focus on the most important and interesting parts of your code. This is a great opportunity to showcase your coding skills and demonstrate your understanding of the project.
- 5) Remember **to be clear and concise** in your explanations and to **highlight any particularly innovative or challenging aspects of your code**. For this task you are encouraged to use **OBS Studio** and you should record both your screen and your voice.
- 6) The video should be **uploaded on YouTube** as unlisted and the link to the video should be shared in a text file and uploaded on VLE as per the **Assignment Guidelines** section of this document.
- 7) A mandatory class presentation must be done using the same workflow of this video. Class presentations will be conducted in front of the whole class and may be followed by technical questions by your lecturer.
- 8) Even if you are unable to fulfill all the application specifications, you are still required to make a video presentation of your work. This is for external verifications purposes.**

Example App Scenarios

The following are examples of mobile application scenarios that can be used for this assignment. Feel free to build on these ideas or go to something completely different. Remember that these scenarios are only for inspiration and to help you with your proposal idea.

1. Task & Productivity Apps

Study Planner App

Features:

- Add new study tasks
- Mark tasks as completed
- Filter by subject or status
- Multiple pages (All Tasks / Add Task / About)

Habit Tracker App

Features:

- Add daily habits
- Toggle completed/not completed
- View today's habits
- Simple statistics page (e.g., completed count)

Simple Budget Tracker

Features:

- Add expense entries
- Show list of expenses
- Display total amount
- Filter by category

2. Lifestyle / Personal Apps

Workout Tracker

Features:

- Add exercises



- Record sets/ reps
- View workout history (list)
- Navigation between 'Add Workout' and 'History'

Recipe Organizer

Features:

- Add recipes
- View recipe list
- Tap to view recipe details (navigation push)

Book / Movie Watchlist

Features:

- Add items
- Mark as watched
- View details page
- Filter watched/unwatched

3. Student-Friendly Utility Apps

Class Notes App

Features:

- Add note
- View list of notes
- Click to see full note
- Delete note

Event Reminder App

Features:

- Add event (title, date, description)
- List events
- Highlight upcoming events
- Navigation menu